

Population Momentum

Why a country can hit replacement fertility and still add tens of millions of people, and what an age pyramid is really telling you.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/population-momentum/

1 Before you open anything

Answer from what you already know. Being right is not the point — the point is to have a number on paper that the projection can later confirm or contradict.

PREDICTION

A country of **100.0 million** people has a wide-based age pyramid: 34% of everyone alive is under 15. **Today** its fertility drops from 3.91 children per woman to replacement, 2.1. Nobody enters or leaves the country. What is its population **70 years from today**? Give a number in millions.

Write one or two sentences on the reasoning that produced that number.

2 Setting up the scenario you just predicted

Open the demonstration. Panel 1 builds the starting pyramid, panel 2 sets the fertility change, and the slider under the pyramid moves you through the projection.

1. Press **Reset everything** in panel 2. Check that panel 1 shows **Fertility that built it = 3.91**, that **New fertility (TFR)** reads **2.10**, that **Change happens at year** is **0**, and that **Phase-in time** reads **instant**.
2. Read the two tiles in panel 1 and write them in the boxes below.
3. Drag **Year of the projection** to each year in the table and copy all four tiles from panel 4, *At the year on the scrubber*. The slider moves in steps of 5, so every year listed lands exactly.

Starting pyramid. Fertility that built it: _____ children per woman. Under age 15: _____ % of everyone alive.

Year	Population (M)	Growth rate (%/yr)	Women aged 15–49 (M)	Median age (yr)
0				
20				
40				
60				
70				
80				

a. The **New fertility (TFR)** slider never moved off 2.10, and this model has no migration at all. So where did the extra people between year 0 and year 60 come from?

b. Fertility is fixed at 2.10 children per woman for the whole run, yet the **Women aged 15–49** column changes a great deal. Births in any five-year step are the fertility rate multiplied by the size of that pool. Use that to explain why total births keep rising after year 0.

c. Between which two years in your table does the growth rate first fall below 0.20 %/yr? Compare that with the year the population stops rising, read off **Growth under 0.1%/yr at** in panel 3.

The population is scaled to start at exactly 100 million so percentages read straight off the numbers. The units are arbitrary; only the ratios carry meaning. Nothing in this model depends on the country being that size.

3 What the pyramid's shape is worth

Now change one thing only. Leave **New fertility (TFR)** at **2.10**, **Change happens at year** at **0** and **Phase-in time** at **instant** for every row. Every starting pyramid here is a genuine stable age structure, so the shape and the fertility that built it always agree before you change anything.

1. Press **Contracting** in panel 1, then fill in that row from panel 1 and panel 3.
2. Press **Stable**, then **Expansive**, filling in those rows.
3. For the two middle rows, drag the **Shape** slider until **Fertility that built it** reads the value in the first column.
4. For the last row, drag **Shape** as far right as it goes.

Fertility that built it	Under age 15 (%)	Growth still to come (%)	95% of it arrives within (yr)
1.76 – <i>Contracting</i>			
2.09 – <i>Stable</i>			
2.79			
3.22			
3.91 – <i>Expansive</i>			
5.16 – <i>slider at maximum</i>			

a. The new fertility was 2.10 in every single row. State the rule relating the *share under age 15* to the growth still to come, in your own words.

b. One row has a negative value in the third column. Say which, and explain what “negative momentum” means for where the large cohorts are sitting in that pyramid.

c. Return to your prediction in part 1. What number did you write, and what does the year-70 row of your part-2 table give? If they differ, name the specific assumption in your reasoning that the projection contradicts.

4 Working backwards, and a shortcut that fails

4A — hit a target. You want a scenario that stores *at least +100%* of growth: a population that doubles after fertility has already reached replacement.

- 1. Press **Reset everything**, then drag **Shape** all the way right. Record **Growth still to come** from panel 3, together with the two population figures printed directly beneath that percentage. Can the shape slider alone reach +100%?
- 2. Press **Expansive** to come back to 3.91. Now raise **Phase-in time** one step at a time until **Growth still to come** first passes +100%.

Setting	Growth still to come (%)	The two figures under it (M)	Growth under 0.1%/yr at (yr)
Shape at maximum, phase-in instant			
Expansive, phase-in ----- yr			

a. A phase-in of several decades is a *slower* fall in fertility, so it might read as a smaller effect. It stores more growth instead. Explain why, in terms of the birth cohorts arriving while fertility is still on its way down.

4B — the shortcut. A common and perfectly reasonable rule of thumb: fertility below 2.1 means the population shrinks. Test it on this pyramid.

- 1. Press **Reset everything**. Set **New fertility (TFR)** to **1.60** — well below replacement. Fill in the first row.
- 2. Leaving 1.60 in place, drag **Year of the projection** to **80** and record the population and growth rate from panel 4.

3. Now set **New fertility (TFR)** to **1.00** — under half of replacement — and fill in the second row.

New fertility	Growth still to come (%)	The two figures under it (M)	Growth under 0.1%/yr at (yr)
1.60			
1.00			

At TFR 1.60, year 80: population _____ M, growth rate _____ %/yr.

b. The rule of thumb is right about the long run and wrong about the next few decades. Rewrite it as a single sentence that is true of both.

5 Solve these without the demonstration

Close the page or look away from it. Show your working. Then reopen it, check yourself, and if a number is off, write down where.

1. A country of **400 million** people has an age pyramid the same shape as the *Expansive* preset, and its fertility falls to replacement this year. Using the stored-growth percentage you recorded in part 3, predict its population when growth stops, and state roughly how many years that takes. Explain in one line why the answer scales this way when the model was built at 100 million.

2. On the *Expansive* run, the pool of women aged 15–49 goes from 22.6 million at year 0 to 33.7 million at year 40, with fertility held at 2.10 throughout. By what factor does the annual number of births change over that period? Then find the fertility that would have been needed at year 40 to hold births at their year-0 level.

3. Replacement fertility in this model is 2.09, not 2.00. Give the two reasons, and state which way the replacement value moves in a country with higher child mortality. No calculation — this one is an explanation.

4. A country whose pyramid matches the *Contracting* preset raises its fertility to replacement, 2.10. Predict whether its population levels off immediately, and give the direction and rough size of the change that follows. Justify it by saying which age groups hold this pyramid's largest cohorts.

6 On your own

Nothing here is graded, and the worksheet has used only part of what the demonstration does.

- The **Change happens at year** slider has not been touched yet. Reset, then move it to 25 and to 50. Watch which readouts in panel 3 change and which do not, and work out why the percentage behaves differently from the two population figures under it.
 - Switch the pyramid between **Share of population (%)** and **People (millions)** and press **Play** in each. The bottom rows do something quite different in the two views. Decide what each view hides.
 - Set **New fertility (TFR)** to 3.00 and read what panel 3 reports instead of a number. Then compare the solid curve with the dashed *if fertility had never changed* line at year 60, and say what the gap between them is measuring.
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